

Exercise 1 (Exam 2018) [2pt. | 2pt. | 4pt.]

Consider the system

$$x(k+1) = Ax(k) + Bu(k)$$

and the standard MPC control law $\kappa(x(k)) = u_0^*(x(k))$ as defined in the supplementary material by optimization problem $P_{S,1}$, satisfying Assumptions S.1-1, S.1-2, S.1-3, with polytopic $\mathcal{X}$, $\mathcal{U}$, $\mathcal{X}_f$ and quadratic stage and terminal cost.

i) Mark the true statements about optimization problem $P_{S,1}$.

	True	False
$P_{S,1}$ is a convex optimization problem.	☐	☐
$P_{S,1}$ is a linear program.	☐	☐
$P_{S,1}$ has a bounded optimal cost.	☐	☐
$P_{S,1}$ has a unique minimizer.	☐	☐

ii) Mark the true statements about the closed-loop system under the MPC control law $\kappa(x(k))$. Assume that $P_{S,1}$ is feasible for $x_0 = x(0)$.

	True	False
$P_{S,1}$ is feasible for all $k \geq 0$.	☐	☐
$x(k) \in \mathcal{X}$ and $u(k) \in \mathcal{U}$ for all $k \geq 0$.	☐	☐
$x(N) \in \mathcal{X}_f$.	☐	☐
$x(k+1)^T Q x(k+1) \leq x(k)^T Q x(k)$.	☐	☐

iii) Show that the infinite-horizon cost of the closed-loop system is always smaller or equal to the predicted open-loop cost, i.e.

$$J_\infty(x(0)) = \sum_{k=0}^{\infty} \|x(k)\|_Q^2 + \|u(k)\|_R^2 \leq J^*(x(0)),$$

for $u(k) = \kappa(x(k))$.

Hint: Remember that in the standard MPC formulation under Assumptions S.1-1, S.1-2, S.1-3, it holds that $J^*(x(k+1)) - J^*(x(k)) \leq -\|x(k)\|_Q^2 - \|u(k)\|_R^2$.

Exercise 2 (Exam 2020) [2pt. | 3pt. | 2pt. | 1pt. | 3pt. | 2pt. | 4pt.]

Consider the following scalar system

$$x(k+1) = 2x(k) + u(k).$$

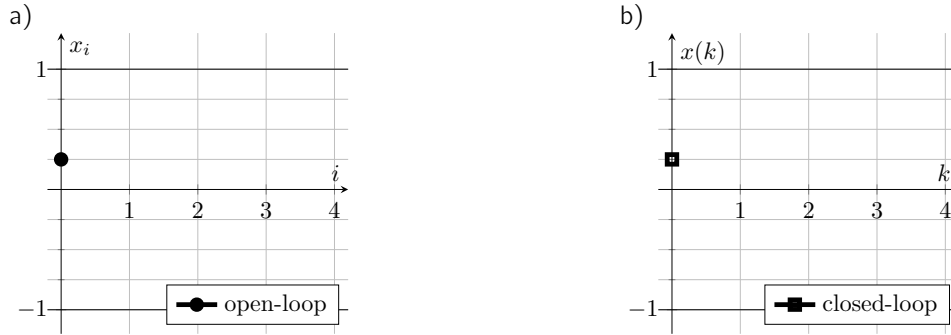
The system is subject to state and input constraints:

$$\begin{aligned} -1 &\leq x \leq 1, \\ -\frac{1}{3} &\leq u. \end{aligned}$$

The goal is to keep the system within the constraints while minimizing the input to the system. A simple MPC control law is proposed

$$\begin{aligned}
 J_{N,1}^*(x(k)) = \underset{X, U}{\text{minimize}} \quad & \sum_{i=0}^{N-1} u_i^2 \\
 \text{subject to} \quad & x_{i+1} = 2x_i + u_i, \quad i = 0, \dots, N-1, \\
 & |x_i| \leq 1, \quad i = 0, \dots, N, \\
 & u_i \geq -\frac{1}{3}, \quad i = 0, \dots, N-1, \\
 & x_0 = x(k),
 \end{aligned} \tag{1}$$

where $N = 2$ and the control law is $\kappa(x(k)) = u_0^*(x(k))$.



- For the initial state $x(0) = 0.25$, provide the optimal open-loop prediction resulting from the MPC problem (1) and plot it in Figure a).
- State the closed-loop system evolution under the MPC controller (1) for two time steps, $\{x(1), x(2)\}$ and plot it in Figure b).
- Provide the feasible set of the MPC problem (1) of the form $\mathcal{X}_{N=2}^{(1)} = \{x | H_N x \leq h_N\}$.
- Is the MPC problem (1) guaranteed to be recursively feasible for any initially feasible state? Justify your answer.

Consider now the modified MPC problem with added terminal constraint $x_N = 0$.

$$\begin{aligned}
 J_{N,2}^*(x(k)) = \underset{X, U}{\text{minimize}} \quad & \sum_{i=0}^{N-1} u_i^2 \\
 \text{subject to} \quad & x_{i+1} = 2x_i + u_i, \quad i = 0, \dots, N-1, \\
 & |x_i| \leq 1, \quad i = 0, \dots, N, \\
 & u_i \geq -\frac{1}{3}, \quad i = 0, \dots, N-1, \\
 & x_N = 0, \\
 & x_0 = x(k),
 \end{aligned} \tag{2}$$

where $N = 2$.

- Provide the feasible set of the MPC problem (2) with horizon $N = 2$ of the form $\mathcal{X}_{N=2}^{(2)} = \{x | H_N x \leq h_N\}$.
- Is the MPC problem (2) with horizon $N = 2$ guaranteed to be recursively feasible? Justify your answer.
- Mark all statements that are guaranteed to be true.

	True
The feasible set of the MPC problem (1) with planning horizon $N=2$ is a subset of the feasible set of the MPC problem (1) with planning horizon $N=3$, i.e., $\mathcal{X}_{N=2}^{(1)} \subseteq \mathcal{X}_{N=3}^{(1)}$.	☐
The feasible set of the MPC problem (2) with planning horizon $N=2$ is a subset of the feasible set of the MPC problem (2) with planning horizon $N=3$, i.e., $\mathcal{X}_{N=2}^{(2)} \subseteq \mathcal{X}_{N=3}^{(2)}$.	☐
For fixed horizon N , the optimal open-loop cost $J_{N,2}^*(x(k))$ of the MPC problem (2) is smaller or equal to the optimal open-loop cost $J_{N,1}^*(x(k))$ of the MPC problem (1) for all $x(k)$ for which both optimization problems are feasible.	☐
For the MPC problem (2) with two different prediction horizons N_1 and N_2 with $N_1 > N_2$ the optimal open-loop cost $J_{N_1,2}^*(x(k))$ is larger or equal to $J_{N_2,2}^*(x(k))$, i.e. $J_{N_1,2}^*(x(k)) \geq J_{N_2,2}^*(x(k))$.	☐

Exercise 3 (Exam 2018) [3pt. | 4pt.]

Consider the following autonomous system with an orthogonal dynamics matrix

$$x(k+1) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} x(k) \quad (3)$$

and state constraints $\mathcal{X}$ as defined in the plots provided below, shown as dashed line.

Hint: For orthogonal matrices it holds that $A^T A = I$.

- Draw the pre-set $\text{pre}(\mathcal{X})$, as well as $\text{pre}(\mathcal{X}) \cap \mathcal{X}$ in Figure a).
- Graphically find the maximum positive invariant set for system (3) under constraints $\mathcal{X}$, and draw it in Figure d). You can use the remaining plots b) - c) for intermediate steps.

